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Thermodynamic assessment of Al–Ce–Cu system

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ABSTRACT

The Ce–Cu binary system has been thermodynamically reassessed by CALPHAD approach based on experimental data and thermodynamic properties of phase diagram. Incorporating the thermodynamic evaluations of Al–Ce and Al–Cu systems from literature, the Al–Ce–Cu ternary system has been assessed using CALPHAD approach. The solution phases including liquid, Fcc and Bcc were treated as substitutional solution phases and the Redlich–Kister polynomial was used to describe their excess Gibbs energies. All binary and ternary intermetallic compounds were treated as stoichiometric phases and their Gibbs energies of formation were optimized. The calculated Al–Ce–Cu phase diagram agreed well with available experimental data.

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1. Introduction

The development of aluminum alloys containing rare earth elements has attracted a lot of attention from material researchers recently [1–7]. One of the main research focuses lies in alloying aluminium–copper based alloys with cerium for improving their castability and mechanical properties. The attempt to use cerium to improve thermal fatigue strength of Al–4.5% Cu alloy under similar conditions to working environment was reported by Si and co-workers [8]. They found that the addition of 0.3 wt.% Ce gave the highest thermal resistance, which is beneficial to mitigate speed of the crack initiation and propagation in alloy without Ce. They attributed their results to (1) Ce atoms segregated at the solidification front of the dendrites so as to increase the region of compositional supercooling and reduce the arm spacing of secondary dendrites, which resulted in finer grains and higher mechanical strength. (2) Partial Ce atoms helped to eliminate the non-metallic inclusion such as Al_2O_3 whose appearance deteriorates the processing and casting properties. According to their work, the amount of the inclusion was reduced as the Ce addition increased up to 0.7 wt.%. (3) A fine crystalline structure was obtained with proper Ce addition (<0.5 wt.%) to give good comprehensive mechanical properties. Similar results were found in Al–Cu–Mg–Ag system with addition of Ce [9]. According to Xiao et al. [9], both hardness and tensile strength were increased for alloys with Ce addition up to 0.45 wt.% which was ascribed to high density and thermal stability of fine Ω precipitates, the major strengthening phase in the Al–Cu–Mg–Ag system.

Since knowledge of phase relations and thermodynamics are fundamental issues to material design, the objective of this work is to develop a thermodynamic description of the Al–Ce–Cu ternary system through CALPHAD approach.

2. Literature review

2.1. The Ce–Cu binary system

The Ce–Cu phase diagram was investigated by Hanaman [10], Gschneidner [11,14], Duisemaliev and Presnyakov [12] and Rhinehammer et al. [13]. They all agreed on the existence of four intermetallic compounds, i.e., Cu_6Ce, Cu_4Ce, Cu_2Ce and CuCe. The Cu_5Ce phase was accepted by most authors [12–14] and also included in this work. The liquidus of the Ce–Cu system obtained by Hanaman [10], Rhinehammer et al. [13] and Gschneidner and Verkade [14] agreed well with each other despite the disagreement on Cu_5Ce.

Duisemaliev and Presnyakov [12] and Korolkov and Lysova [15] have reported the maximum solid solubility of Ce in Cu which is close to 0.1 at.% Ce on the basis of metallography, microhardness and resistivity data. The maximum solid solubilities of Cu in γCe and δCe are around 0.37 and 0.55 at.% Cu, respectively [11].

The partial enthalpies of Cu and Ce in their binary alloys at 1523 K were measured with a high-temperature calorimeter by Turchanin et al. [16]. Fitzner and Kleppa [17] measured the enthalpies of mixing of liquid Ce–Cu alloys by high-temperature calorimeter at

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Table 1
Thermodynamic parameters of the Al–Ce–Cu system.

Phase	Thermodynamic parameters	Ref.
Liquid model: (Al,Ce,Cu)	${}^{0}L^{Liq}_{Al,Ce} = -130371 + 38.523T$	[25]
	${}^{1}L^{Liq}_{Al,Ce} = -36330 + 6.362T$	[25]
	${}^{2}L^{Liq}_{Al,Ce} = -9208 + 4.065T$	[25]
	${}^{0}L^{Liq}_{Al,Cu} = -67094 + 8.555T$	[23]
	${}^{1}L^{Liq}_{Al,Cu} = +32148 - 7.118T$	[23]
	${}^{2}L^{Liq}_{Al,Cu} = +5915 - 5.889T$	[23]
	${}^{3}L^{Liq}_{Al,Cu} = -8175 + 6.049T$	[23]
	${}^{0}L^{Liq}_{Ce,Cu} = -44488 - 0.398T$	This work
	${}^{1}L^{Liq}_{Ce,Cu} = +34088 - 6.247T$	This work
	${}^{2}L^{Liq}_{Ce,Cu} = -20951 + 10.387T$	This work
	${}^{0}L^{Liq}_{Al,Ce,Cu} = -130000$	This work
Fcc model: (Al,Ce,Cu)	${}^{0}L^{Fcc}_{Al,Ce} = -62851 + 30.556T$	[25]
	${}^{0}L^{Fcc}_{Al,Cu} = -53520 + 2T$	[23]
	${}^{1}L^{Fcc}_{Al,Cu} = +38590 - 2T$	[23]
	${}^{2}L^{Fcc}_{Al,Cu} = 1170$	[23]
	${}^{0}L^{Fcc}_{Ce,Cu} = 25019$	This work
Bcc model: (Al,Ce,Cu)	${}^{0}L^{Bcc}_{Al,Ce} = -93683 + 37.261T$	[25]
	${}^{0}L^{Bcc}_{Al,Cu} = -73554 + 4T$	[23]
	${}^{1}L^{Bcc}_{Al,Cu} = +51500 - 11.84T$	[23]
	${}^{0}L^{Bcc}_{Ce,Cu} = -8545$	This work
Al_4Ce model: $(Al)_{0.8}(Ce)_{0.2}$	$G^{Al_4Ce}_{Al:Ce} = -29557 + 2.001T + 0.8GHSERAL + 0.2GHSERCE$	[25]
$Al_{11}Ce_3$ model: $(Al)_{0.7857}(Ce)_{0.2143}$	$G^{Al_{11}Ce_3}_{Al:Ce} = -42745 + 11.159T + 0.7857GHSERAL + 0.2143GHSERCE$	[25]
βAl_3Ce model: $(Al)_{0.75}(Ce)_{0.25}$	$G^{\beta Al_3Ce}_{Al:Ce} = -44437 + 9.867T + 0.75GHSERAL + 0.25GHSERCE$	[25]
αAl_3Ce model: $(Al)_{0.75}(Ce)_{0.25}$	$G^{\alpha Al_3Ce}_{Al:Ce} = -45421 + 10.658T + 0.75GHSERAL + 0.25GHSERCE$	[25]
Al_2Ce model: $(Al)_{0.6667}(Ce)_{0.3333}$	$G^{Al_2Ce}_{Al:Ce} = -50060 + 9.889T + 0.6667GHSERAL + 0.3333GHSERCE$	[25]
AlCe model: $(Al)_{0.5}(Ce)_{0.5}$	$G^{AlCe}_{Al:Ce} = -48000 + 14.468T + 0.5GHSERAL + 0.5GHSERCE$	[25]
$AlCe_2$ model: $(Al)_{0.3333}(Ce)_{0.6667}$	$G^{AlCe_2}_{Al:Ce} = -24816 + 0.714T + 0.3333GHSERAL + 0.6667GHSERCE$	[25]
$\beta AlCe_3$ model: $(Al)_{0.25}(Ce)_{0.75}$	$G^{\beta AlCe_3}_{Al:Ce} = -27935 + 9.816T + 0.25GHSERAL + 0.75GHSERCE$	[25]
$\alpha AlCe_3$ model: $(Al)_{0.25}(Ce)_{0.75}$	$G^{\alpha AlCe_3}_{Al:Ce} = -28226 + 10.372T + 0.25GHSERAL + 0.75GHSERCE$	[25]
θ Model: $(Al)_2(Al,Cu)_1$	$G^{\theta}_{Al:Al} = +30249 - 14.439T + 3GHSERAL$	[23]
	$G^{\theta}_{Al:Cu} = -47406 + 6.75T + 2GHSERAL + GHSERCU$	[23]
	${}^{0}L^{\theta}_{Al:Al,Cu} = 2211$	[23]
η Model: $(Al,Cu)_1(Cu)_1$	$G^{\eta}_{Al:Cu} = -40560 + 3.14T + GHSERAL + GHSERCU$	[23]
	$G^{\eta}_{Cu:Cu} = +8034 - 2.51T + 2GHSERCU$	[23]
	${}^{0}L^{\eta}_{Al,Cu:Cu} = -25740 - 20T$	[23]
ε Model: $(Al,Cu)_1(Cu)_1$	$G^{\varepsilon}_{Al:Cu} = -36976 + 1.2T + GHSERAL + GHSERCU$	[23]
	$G^{\varepsilon}_{Cu:Cu} = +8034 - 2.51T + 2GHSERCU$	[23]
	${}^{0}L^{\varepsilon}_{Al,Cu:Cu} = +7600 - 24T$	[23]
	${}^{1}L^{\varepsilon}_{Al,Cu:Cu} = -72000$	[23]
ζ Model: $(Al)_9(Cu)_{11}$	$G^{\zeta}_{Al:Cu} = -420000 + 18T + 9GHSERAL + 11GHSERCU$	[23]
δ Model: $(Al)_2(Cu)_3$	$G^{\delta}_{Al:Cu} = -106700 + 3T + 2GHSERAL + 3GHSERCU$	[23]
γ_D8$_3$ Model: $(Al)_4(Al,Cu)_1(Cu)_8$	$G^{\gamma_D8_3}_{Al:Al:Cu} = -277739 + 215T - 30T\ln(T) + 5GHSERAL + 8GHSERCU$	[23]
	$G^{\gamma_D8_3}_{Al:Cu:Cu} = -280501 + 379.6T - 52T\ln(T) + 4GHSERAL + 9GHSERCU$	[23]
γ Model: $(Al)_4(Al,Cu)_1\ (Cu)_8$	$G^{\gamma}_{Al:Al:Cu} = -219258 - 45.5T + 5GHSERAL + 8GHSERCU$	[23]
	$G^{\gamma}_{Al:Al:Cu} = -200460 - 58.5T + 4GHSERAL + 9GHSERCU$	[23]
Cu_6Ce model: $(Cu)_{0.857}(Ce)_{0.143}$	$G^{Cu_6Ce}_{Cu:Ce} = -8964 - 2.05T + 0.857GHSERCU + 0.143GHSERCE$	This work
Cu_5Ce model: $(Al,Cu)_{0.833}(Ce)_{0.167}$	$G^{Cu_5Ce}_{Cu:Ce} = -10290 - 1.5T + 0.833GHSERCU + 0.167GHSERCE$	This work
	$G^{Cu_5Ce}_{Al:Ce} = -17260 + 0.8T + 0.833GHSERAL + 0.167GHSERCE$	This work
	${}^{0}L^{Cu_5Ce}_{Al,Cu:Ce} = -66700 + 0.6T$	This work
Cu_4Ce model: $(Al,Cu)_{0.8}(Ce)_{0.2}$	$G^{Cu_4Ce}_{Cu:Ce} = -11640 - 1.174T + 0.8GHSERCU + 0.2GHSERCE$	This work
	$G^{Cu_4Ce}_{Al:Ce} = -22000 + 0.7T + 0.8GHSERAL + 0.2GHSERCE$	This work
	${}^{0}L^{Cu_4Ce}_{Al,Cu:Ce} = -65000 + 0.7T$	This work
Cu_2Ce model: $(Cu)_{0.667}(Ce)_{0.333}$	$G^{Cu_2Ce}_{Ce,Cu} = -11440 - 4.541T + 0.667GHSERCU + 0.333GHSERCE$	This work

Table 1 (*Continued*)

Phase	Thermodynamic parameters	Ref.
CuCe model: $(Cu)_{0.5}(Ce)_{0.5}$	$G^{CuCe}_{Ce,Cu} = -9420 - 3.5T + 0.5GHSERCU + 0.5GHSERCE$	This work
τ_1 Model: $(Al)_{0.6154}(Ce)_{0.0769}(Cu)_{0.3077}$	$G^{\tau_1}_{Al:Ce:Cu} = -30000 + 3T + 0.6154GHSERAL + 0.0769GHSERCE + 0.3077GHSERCU$	This work
τ_2 Model: $(Al,Cu)_{0.8947}(Ce)_{0.1053}$	$G^{\tau_2}_{Al:Ce} = -13200 + 2.5T + 0.8947GHSERAL + 0.1053GHSERCE$	This work
	$G^{\tau_2}_{Cu:Ce} = -4000 + 0.8T + 0.8947GHSERCU + 0.1053GHSERCE$	This work
	${}^0L^{\tau_2}_{Al,Cu:Ce} = -75000 + 0.8T$	This work
τ_3 Model: $(Al,Cu)_{0.8}(Ce)_{0.2}$	$G^{\tau_3}_{Al:Ce} = -29557 + 2.1T + 0.8GHSERAL + 0.2GHSERCE$	This work
	$G^{\tau_3}_{Cu:Ce} = -2000 + 0.3T + 0.8GHSERCU + 0.2GHSERCE$	This work
	${}^0L^{\tau_3}_{Al,Cu:Ce} = -68500 + 0.4T$	This work
τ_4 Model: $(Al)_{0.3333}(Ce)_{0.3333}(Cu)_{0.3334}$	$G^{\tau_4}_{Al:Ce:Cu} = -34600 + 0.4T + 0.3333GHSERAL + 0.3333GHSERCE + 0.3334GHSERCU$	This work
τ_5 Model: $(Al)_{0.2}(Ce)_{0.4}(Cu)_{0.4}$	$G^{\tau_5}_{Al:Ce:Cu} = -30000 + 0.2T + 0.2GHSERAL + 0.4GHSERCE + 0.4GHSERCU$	This work

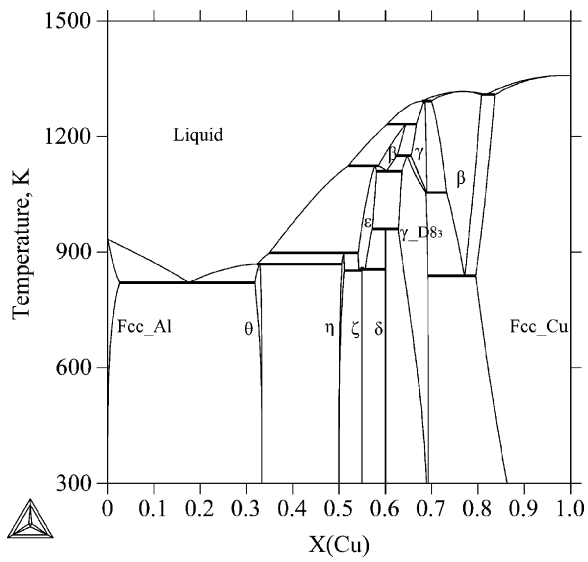

Fig. 1. The Al–Cu phase diagram calculated by Witusiewicz et al. [23].

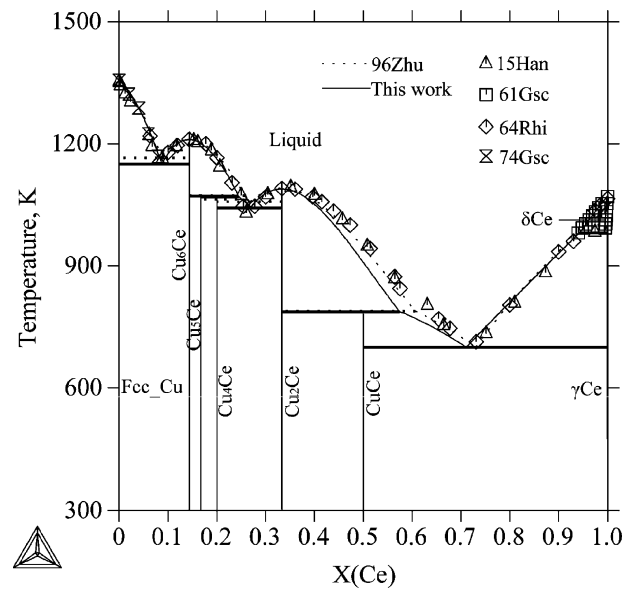

Fig. 3. Calculated phase diagram of the Ce–Cu system.

1473 K using untreated commercial copper powder. However, there was a very large discrepancy between their results and those of Turchanin et al. [16]. For this reason, Fitzner and Kleppa re-determined the enthalpies of mixing of liquid at with either small pieces of copper wire or copper tablets sintered in a stream of hydrogen gas at about 873 K [18]. The enthalpies of mixing of liquid measured in this way were in good agreement with Turchanin et al.'s data. Moreover, the standard enthalpies of formation of two congruently

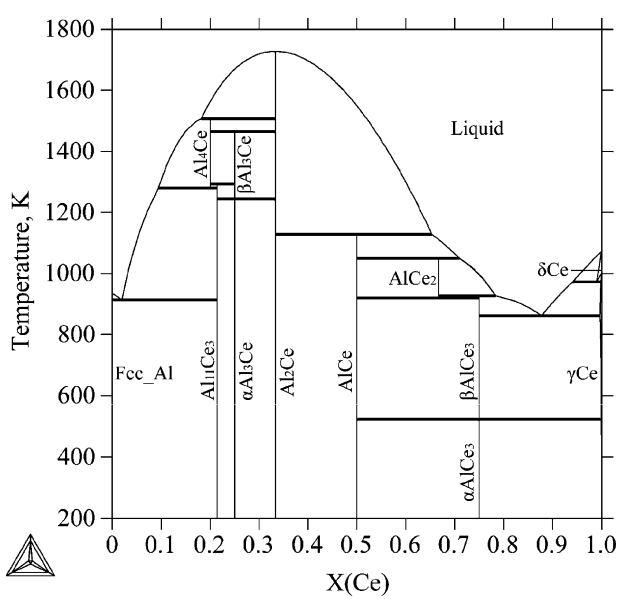

Fig. 2. The Al–Ce phase diagram calculated by Gao et al. [25].

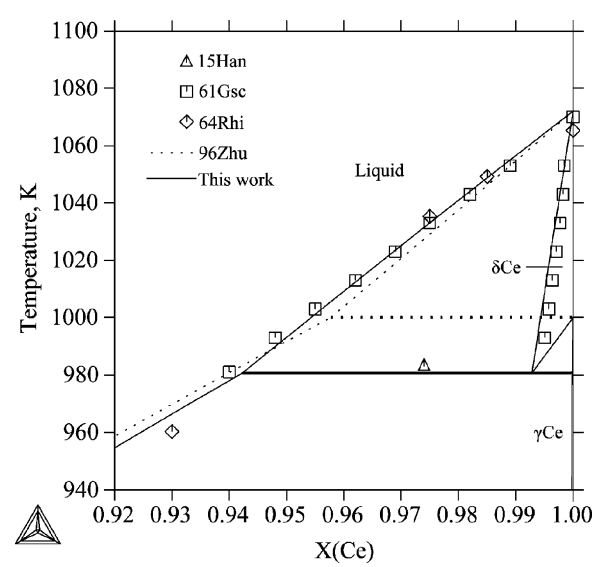

Fig. 4. Calculated enlarged portion of the cerium-rich region.

Table 2
Invariant reactions in the Ce–Cu system.

Reaction	Temperature (K)	Composition of liquid (X_{Ce})	Type	Ref.
Liquid ↔ Fcc_Cu + Cu_6Ce	1149	0.09	Eutectic	[13]
	1165	0.088		[19]
	1150	0.087		This work
Liquid ↔ Cu_6Ce	1211	0.143	Congruent	[13]
	1211	0.143		[19]
	1210	0.143		This work
Liquid + Cu_6Ce ↔ Cu_5Ce	1071	0.234	Peritectic	[13]
	1072	0.257		[19]
	1071	0.244		This work
Liquid + Cu_5Ce ↔ Cu_4Ce	1069	0.235	Peritectic	[13]
	1063	0.263		[19]
	1069	0.246		This work
Liquid ↔ Cu_4Ce + Cu_2Ce	1029	0.260	Eutectic	[13]
	1057	0.268		[19]
	1043	0.269		This work
Liquid ↔ Cu_2Ce	1090	0.333	Congruent	[13]
	1090	0.333		[19]
	1089	0.333		This work
Liquid + Cu_2Ce ↔ CuCe	789	0.600	Peritectic	[13]
	788	0.606		[19]
	787	0.574		This work
Liquid ↔ CuCe + γCe	697	0.720	Eutectic	[13]
	700	0.730		[19]
	700	0.711		This work
δCe ↔ liquid + γCe	981	0.940	Catatectic	[13]
	999	0.958		[19]
	981	0.942		This work

melted solid phases including Cu_6Ce and Cu_2Ce were obtained from the measured heats of reaction at 1473 K and the heat contents of the reaction products at the same temperature [18].

Zhuang et al. [19] have evaluated the Ce–Cu system thermodynamically before. Experimental phase diagram data [10,11,13,14], enthalpies of mixing of liquid alloys [16,18] and heat contents [20,21] were used in their optimization. Although the calculated results agreed well with most of the experimental data, the largest deviation of invariant reaction temperature was 28 K. In addition, Zhuang et al. [19] did not consider the solubility of Cu in γCe and δCe and that of Ce in Fcc_Cu. Therefore, the Ce–Cu system was reassessed in this work.

2.2. The Al–Cu and Al–Ce binary system

The Al–Cu system was thermodynamically assessed by Saunders et al. [22]. Recently, Witusiewicz et al. [23] reassessed the liquid and γ_$D8_3$ phases on the basis of new experimental thermodynamic data. In this work, we adopted the thermodynamic parameters of Witusiewicz et al. [23] and the calculated phase diagram of Al–Cu system is shown in Fig. 1.

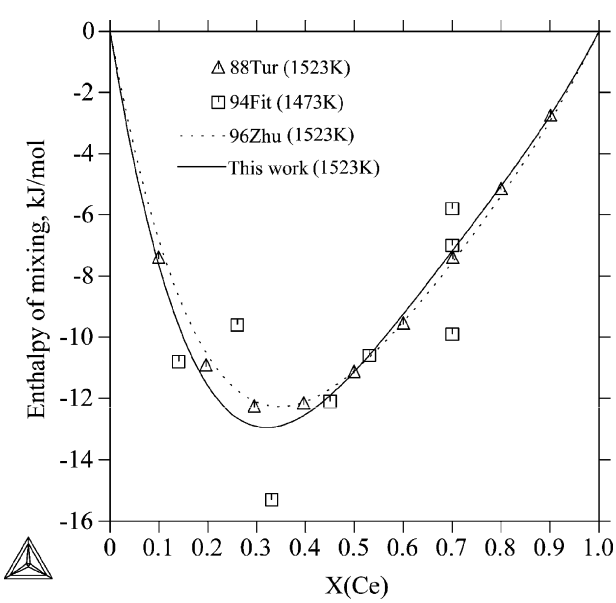

Fig. 5. Calculated enthalpies of mixing of the liquid Ce–Cu alloys at 1523 K.

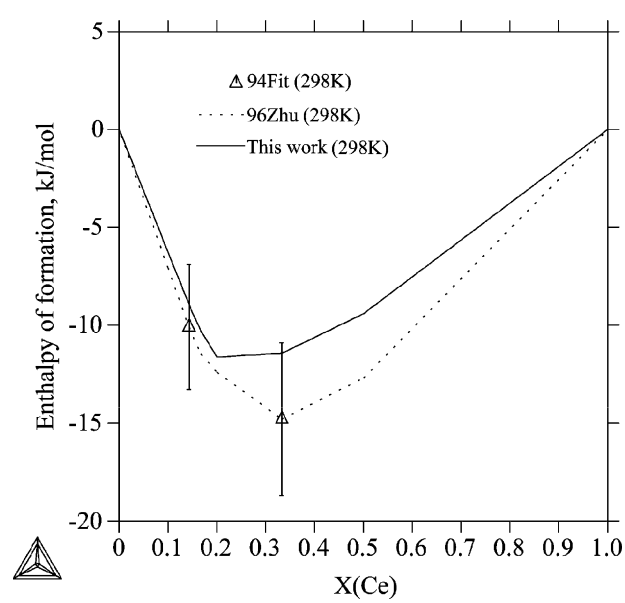

Fig. 6. Calculated standard enthalpies of formation for solid Ce–Cu alloys.

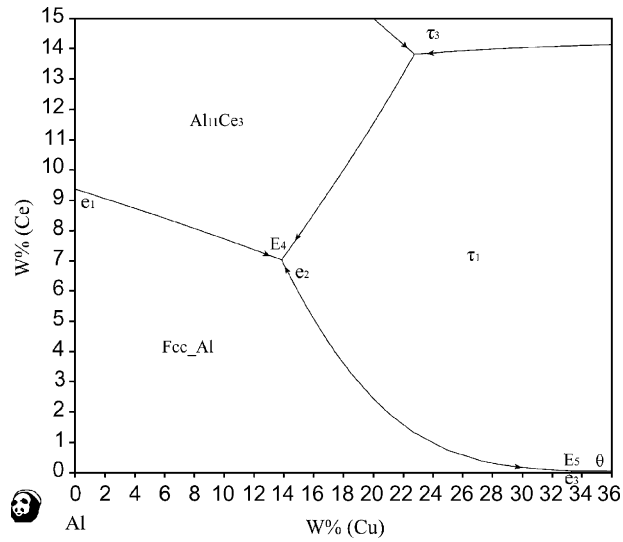

Fig. 7. Calculated liquidus projection of the Al-rich part.

The Al–Ce phase diagram was assessed by Cacciamani and Ferro [24]. Gao et al. [25] reinvestigated this system a few years later, assisted by first-principle calculations and key experiments. Both experimental results and theoretical calculations showed that Al_2Ce should be treated as a stoichiometric phase rather than the solution phase proposed in previous studies. And a new compound, $AlCe_2$, was found stable at high temperatures (648–775 °C). Further, the previously described $\beta Al_{11}Ce_3$ phase was proposed to have a stoichiometry of Al_4Ce, evidenced by differential scanning calorimetry (DSC) measurements. The thermodynamic description of Al–Ce system by Gao et al. [25] adopted in this study is shown in Fig. 2.

2.3. The Al–Ce–Cu ternary system

Limited information of phase diagram exists for the Al–Ce–Cu ternary system. It was Yunusov et al. [26] who first investigated the Al-rich corner. One liquidus projection and four quasi-binary sections such as Al–Al_8CeCu_4, Al–Al_4CeCu, Al_8CeCu_4–Al_4CeCu and Al_4CeCu–$CeAl_2$ were reported. Subsequently, literature [27] reported a liquidus projection of the Al-rich part which was different from that of Yunusov et al. [26], two isothermal sections at 400 and 500 °C and four quasi-binary sections based on the results from Yunusov et al. [26]. More recently, Belov et al. [28,29] studied the phase diagram of the Al-rich corner twice in the consecutive two years. The former included Al_4CeCu phase and the latter not. In this work, we adopted the new results from Belov and Khvan [29], in which the Al_4CeCu phase was supposed to be Al_4Ce phase with Cu being substituted by Al. By using these results, the Al_4CeCu phase in the phase equilibria in the Al-rich corner should be considered as the Al_4Ce phase.

According to the literature [27], there are five ternary compounds in the Al–Ce–Cu phase diagram, which are Al_8CeCu_4, $(Al,Cu)_{17}Ce_2$, $(Al,Cu)_4Ce$, AlCeCu and $Al_{6.5}CeCu_{6.5}$. As the $Al_{6.5}CeCu_{6.5}$ phase was not reported in isothermal sections at 400 and 500 °C, we did not include this compound in this work. For the AlCeCu phase, Chevalier and Bobet [30] suggested that it decomposes at a temperature between 600 and 650 °C peritectically with the formation of $(Cu_{5-x}Al_x)Ce$, Al_2Ce and a new ternary compound corresponding to the $AlCe_2Cu_2$ composition. Therefore, $AlCe_2Cu_2$ phase was included for modeling. Of all the ternary compounds considered in this work, τ_1-Al_8CeCu_4 and τ_4-AlCeCu are strict stoichiometric phases, while τ_2-$(Al,Cu)_{17}Ce_2$ and τ_3-$(Al,Cu)_4Ce$ are generalized stoichiometric phases since Cu could be substituted by Al to some degree. Without any information on the structure, we treated τ_5-$AlCe_2Cu_2$ as a strict stoichiometric phase.

Table 3
Calculated invariant reactions in the Al–Ce–Cu ternary system.

Type	Reaction	T (°C)
E1	Liquid ↔ Cu_5Ce + β + Fcc_Cu	832
E2	τ_2 ↔ liquid + τ_3 + τ_1	828
E3	Liquid ↔ Cu_5Ce + Fcc_Cu + Cu_6Ce	815
E4	Liquid ↔ $Al_{11}Ce_3$ + τ_1 + Fcc_Al	602
E5	Liquid ↔ τ_1 + Fcc_Al + θ	547
E6	Liquid ↔ τ_5 + $\beta AlCe_3$ + γCe	412
E7	Liquid ↔ γCe + CuCe + τ_5	368
U1	Liquid + Al_2Ce ↔ τ_3 + Al_4Ce	1230
U2	τ_3 + Al_4Ce ↔ liquid + $Al_{11}Ce_3$	1007
U3	Liquid + Cu_4Ce ↔ Cu_5Ce + τ_3	974
U4	Liquid + τ_3 ↔ Cu_4Ce + Al_2Ce	974
U5	Liquid + τ_3 ↔ τ_2 + Cu_5Ce	907
U6	Liquid + γ ↔ Cu_5Ce + β	852
U7	Liquid + γ ↔ γ_$D8_3$ + Cu_5Ce	844
U8	Liquid + Cu_5Ce ↔ τ_2 + γ_$D8_3$	838
U9	Liquid + β ↔ γ_$D8_3$ + ε	836
U10	Liquid + γ_$D8_3$ ↔ ε+τ_2	818
U11	Liquid + τ_2 ↔ τ_1 + ε	783
U12	Liquid + τ_3 ↔ $Al_{11}Ce_3$ + τ_1	726
U13	Liquid + $AlCe_2$ ↔ AlCe + $\beta AlCe_3$	647
U14	Liquid + ε ↔ τ_1 + η	625
U15	Liquid + Al_2Ce ↔ τ_4 + AlCe	619
U16	Liquid + Cu_4Ce ↔ τ_5 + τ_4	617
U17	Liquid + η ↔ τ_1 + θ	596
U18	Liquid + τ_4 ↔ AlCe + τ_5	593
U19	Liquid + Cu_4Ce ↔ τ_5 + Cu_2Ce	556
U20	Liquid + AlCe ↔ τ_5 + $\beta AlCe_3$	482
U21	Liquid + Cu_2Ce ↔ CuCe + τ_5	437
P1	Liquid + γ+β ↔ γ_$D8_3$	877
P2	Liquid + Cu_4Ce + Al_2Ce ↔ τ_4	633

Table 4
Calculated invariant reactions in the Al-rich part compared with the experimental data.

Type	Reaction	Concentrations in liquid		T (°C)	Ref.
		Cu (wt.%)	Ce (wt.%)		
e_1	Liquid ↔ Fcc_Al + $Al_{11}Ce_3$	–	9.8	645	[29]
		–	9.4	640	This work
e_2	Liquid ↔ Fcc_Al + τ_1	14	7	610	[29]
		13.8	7.0	602	This work
e_3	Liquid ↔ Fcc_Al + θ	33	–	548	[29]
		33.3	–	547	This work
E_4	Liquid ↔ $Al_{11}Ce_3$ + τ_1 + Fcc_Al	13–14	~7	~605	[29]
		13.8	7.0	602	This work
E_5	Liquid ↔ τ_1 + Fcc_Al + θ	32	~0.5	545	[29]
		33.2	0.05	547	This work

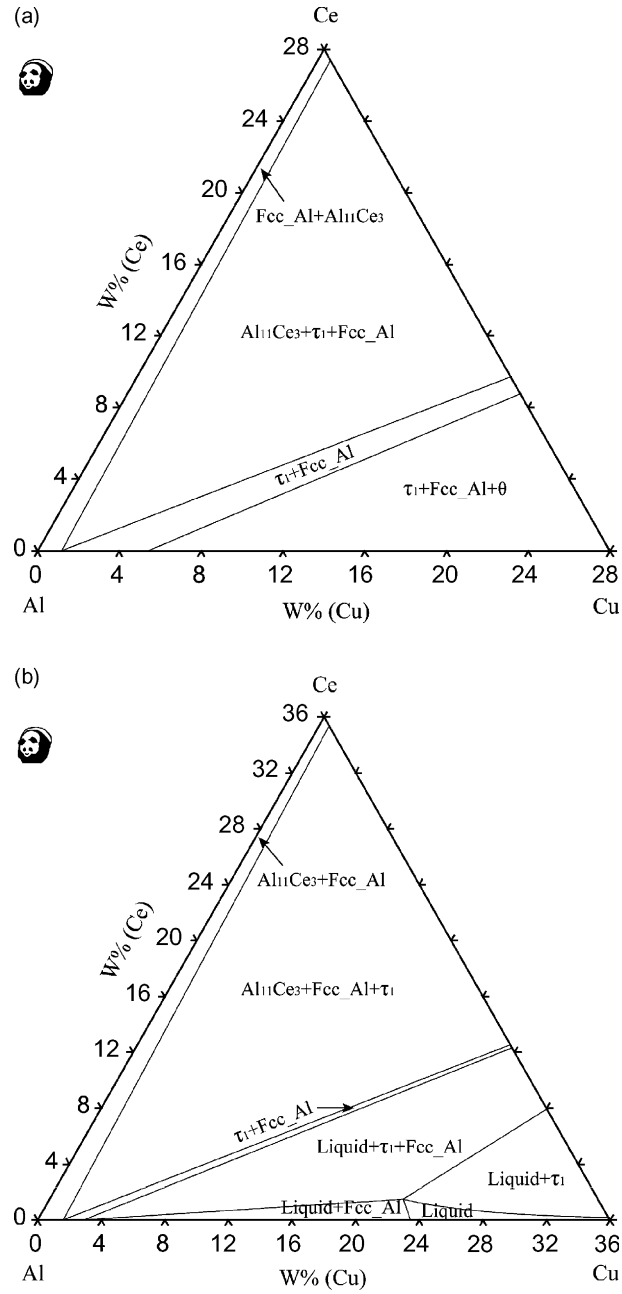

Fig. 8. Calculated isothermal sections at 540 °C (a) and 590 °C (b).

The crystallographic data of all the solid phases can be found in literature [27].

3. Thermodynamic model

3.1. Solution phases

Substitutional solution model is used to describe all the solution phases including liquid, Fcc and Bcc. The mole Gibbs energy can be expressed as a sum of Gibbs energy of pure elements, Gibbs energy contributed from ideal entropy of mixing and excess Gibbs energy, i.e.,

$$G^{\phi} = \sum_{i=\mathrm{Al,Ce,Cu}} x_i^0 G_i^{\phi} + RT \sum_{i=\mathrm{Al,Ce,Cu}} x_i \ln(x_i) + {}^{\mathrm{ex}}G^{\phi} \quad (1)$$

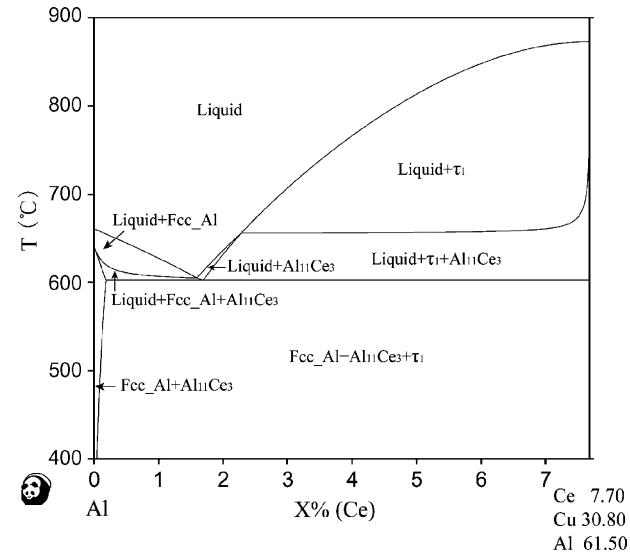

Fig. 9. Calculated quasi-binary section of Al-τ_1.

where ϕ denotes the solution phases, x_i denotes the mole fraction of component i (i = Al, Ce and Cu), and ${}^0G_i^{\phi}$ is the molar Gibbs energy of pure element i in the structure of phase ϕ, which are taken from the SGTE data compiled by Dinsdale [31]. The excess term is the function of the concentration, represented by the Redlich–Kister polynomial, i.e.,

$$\begin{aligned} {}^{\mathrm{ex}}G^{\phi} = {} & x_{\mathrm{Al}}x_{\mathrm{Ce}} \sum_{j=0,1\ldots}^{N} (x_{\mathrm{Al}} - x_{\mathrm{Ce}})^{j(j)} L_{\mathrm{Al,Ce}}^{\phi} \\ & + x_{\mathrm{Al}}x_{\mathrm{Cu}} \sum_{j=0,1\ldots}^{N} (x_{\mathrm{Al}} - x_{\mathrm{Cu}})^{j(j)} L_{\mathrm{Al,Cu}}^{\phi} \\ & + x_{\mathrm{Ce}}x_{\mathrm{Cu}} \sum_{j=0,1\ldots}^{N} (x_{\mathrm{Ce}} - x_{\mathrm{Cu}})^{j(j)} L_{\mathrm{Ce,Cu}}^{\phi} + x_{\mathrm{Al}}x_{\mathrm{Ce}}x_{\mathrm{Cu}} L_{\mathrm{Al,Ce,Cu}}^{\phi} \end{aligned} \quad (2)$$

${}^{(j)}L_{\mathrm{Al,Ce}}^{\phi}$ is taken from Gao et al. [25] and ${}^{(j)}L_{\mathrm{Al,Cu}}^{\phi}$ is taken from Witusiewicz et al. [23]. ${}^{(j)}L_{\mathrm{Ce,Cu}}^{\phi}$ is expressed as

$${}^{(j)}L_{\mathrm{Ce,Cu}}^{\phi} = A_j + B_j T \quad (3)$$

where A_j and B_j are the parameters to be optimized.

$L_{\mathrm{Al,Ce,Cu}}^{\phi}$ are ternary interaction parameters, which are set to zero for Fcc and Bcc phases for lack of experimental data. ${}^0L_{\mathrm{Al,Ce,Cu}}^{\mathrm{liq}}$ is given a negative number for a better fit with the experimental data of the Al-rich part.

3.2. Binary intermetallic phases

All the binary intermetallic compounds are treated as stoichiometric phases, mostly denoted as Cu_xCe_y. According to Neumann–Kopp rule, the Gibbs energy of the Cu_xCe_y phase is represented as

$$G_{\mathrm{Cu}_x\mathrm{Ce}_y} = \frac{x}{x+y}\,{}^0G_{\mathrm{Cu}}^{\mathrm{Fcc}} + \frac{y}{x+y}\,{}^0G_{\mathrm{Ce}}^{\mathrm{Fcc}} + A + BT \quad (4)$$

where A and B are the parameters to be optimized in this work. A is related to enthalpy of formation of the intermetallic compound Cu_xCe_y and A + BT is considered as Gibbs energy of formation of the compound.

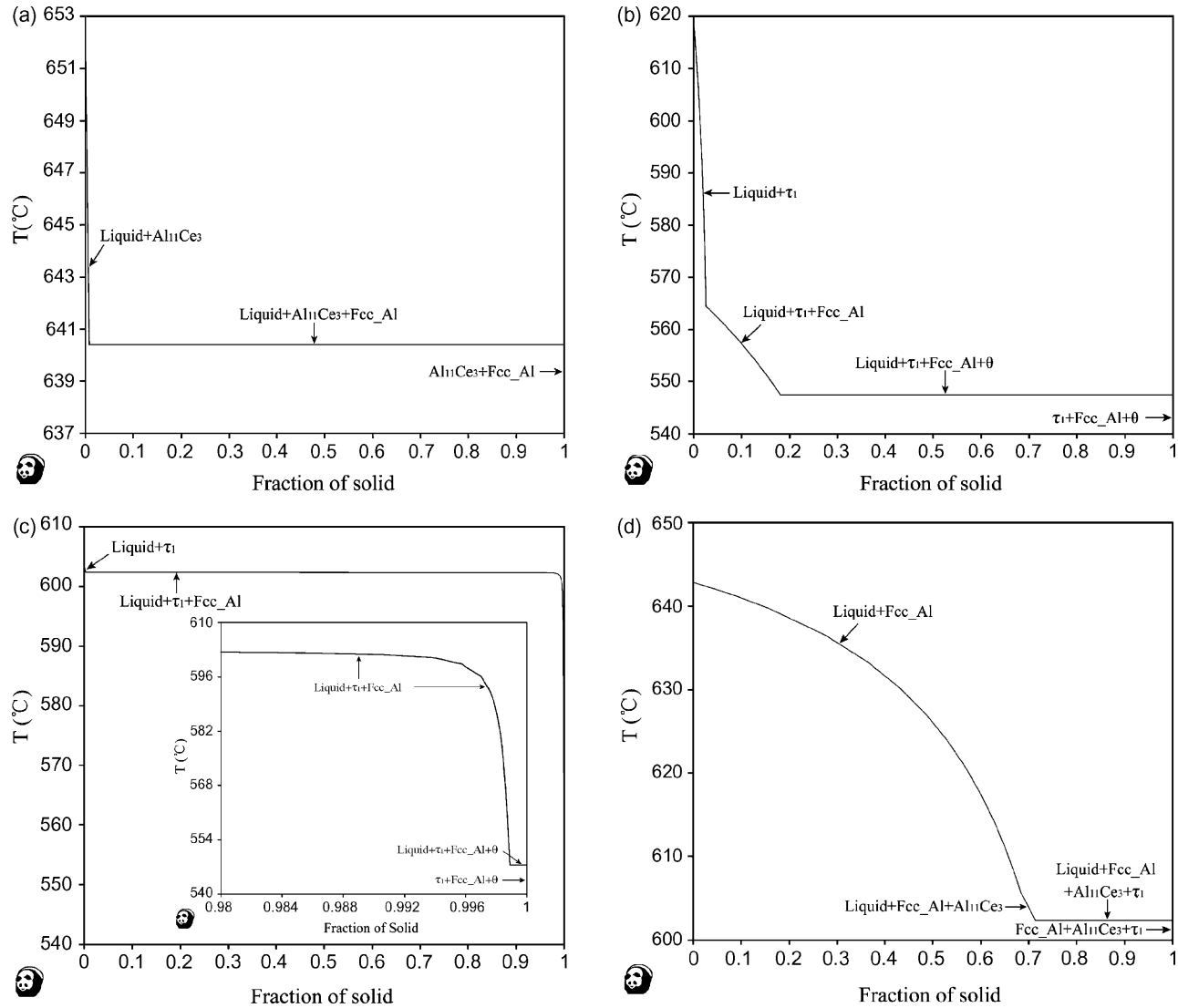

Fig. 10. Simulations of non-equilibrium solidification with Scheil model: (a) 0 wt.% Cu, 10 wt.% Ce; (b) 30 wt.% Cu, 1 wt.% Ce; (c) 14 wt.% Cu, 7 wt.% Ce; (d) 5 wt.% Cu, 2.5 wt.% Ce.

The Cu_4Ce and Cu_5Ce phases are modeled as $(Al,Cu)_xCe_y$ since they have a homogeneity region in the Al–Ce–Cu system according to the literature [27]. Their Gibbs energies are expressed as

$$G_{(Al,Cu)_xCe_y} = y'_{Al}G_{Al:Ce} + y'_{Cu}G_{Cu:Ce} + \frac{x}{x+y}RT(y'_{Al}\ln y'_{Al} + y'_{Cu}\ln y'_{Cu}) + y'_{Al}y'_{Cu}\left(\sum_{j=0,1\ldots} {}^{j}L_{Al,Cu:Ce}(y'_{Al} - y'_{Cu})^{j}\right) \quad (5)$$

where $G_{Cu:Ce}$ has the same value as $G^{Cu_4Ce}_{Cu:Ce}$ or $G^{Cu_5Ce}_{Cu:Ce}$, and $G_{Al:Ce}$ represents Gibbs energy of the Al_4Ce with Cu_4Ce structure or Al_5Ce with Cu_5Ce structure which is formalized as

$$G_{Al:Ce} = \frac{x}{x+y}{}^{0}G^{Fcc}_{Al} + \frac{y}{x+y}{}^{0}G^{Fcc}_{Ce} + A + BT \quad (6)$$

where A and B are the parameters to be optimized in this work.

3.3. Ternary intermetallic compounds

In the Al–Ce–Cu ternary system, strict stoichiometric phases τ_1-Al_8CeCu_4, τ_4-AlCeCu and τ_5-$AlCe_2Cu_2$ are modeled as $Al_xCe_yCu_z$, while generalized stoichiometric phases τ_2-$(Al,Cu)_{17}Ce_2$ and τ_3-$(Al,Cu)_4Ce$ are modeled as $(Al,Cu)_xCe_y$. The Gibbs energy expressions for these two kinds of phases are expressed as

$$G_{Al_xCe_yCu_z} = \frac{x}{x+y+z}{}^{0}G^{Fcc}_{Al} + \frac{y}{x+y+z}{}^{0}G^{Fcc}_{Ce} + \frac{z}{x+y+z}{}^{0}G^{Fcc}_{Cu} + A + BT \quad (7)$$

$$G_{(Al,Cu)_xCe_y} = y'_{Al}G_{Al:Ce} + y'_{Cu}G_{Cu:Ce} + \frac{x}{x+y}RT(y'_{Al}\ln y'_{Al} + y'_{Cu}\ln y'_{Cu}) + y'_{Al}y'_{Cu}\left(\sum_{j=0,1\ldots} {}^{j}L_{Al,Cu:Ce}(y'_{Al} - y'_{Cu})^{j}\right) \quad (8)$$

where

$$G_{Al:Ce} = \frac{x}{x+y}{}^{0}G^{Fcc}_{Al} + \frac{y}{x+y}{}^{0}G^{Fcc}_{Ce} + A + BT \quad (9)$$

$$G_{Cu:Ce} = \frac{x}{x+y}{}^{0}G^{Fcc}_{Cu} + \frac{y}{x+y}{}^{0}G^{Fcc}_{Ce} + A + BT \quad (10)$$

$L_{Al,Cu:Ce}$ is the interaction parameter between Al and Cu in the first sublattice.

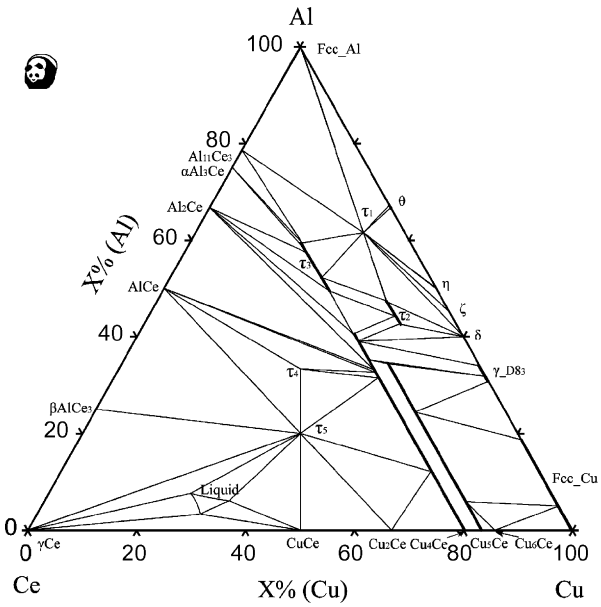

Fig. 11. Calculated isothermal section at 400 °C.

4. Optimization results and discussion

4.1. The Ce–Cu binary system

The optimization of the Ce–Cu system was performed by Parrot module in the Thermo_calc program developed by Sundman et al. [32]. Also, Pandat software [33] was used to check our database. The phase diagram data, the enthalpies of mixing of liquid and the enthalpies of formation of the congruently melted intermetallic compounds were used for optimization. A certain weight was given for each piece of selected information by personal judgment, and during the assessment a trial and error method was applied for verification until most of the selected experimental information was reproduced well. All the parameters optimized in this work are listed in Table 1.

Fig. 3 shows the calculated phase diagram of the Ce–Cu system compared with the work from Zhuang et al. [19] and experimental data. Table 2 lists the calculated invariant reactions in comparison with the results from Rhinehammer et al. [13] and Zhuang et al. [19]. Almost all the calculated temperatures agree very well with the experimental data within the range of 3 K as shown in the enlarged potion of the cerium-rich region in Fig. 4. Even though the calculated temperature for the eutectic reaction $L \leftrightarrow Cu_4Ce + Cu_2Ce$ is 14 K higher than the experimental data, the difference is smaller than the 28 K reported by Zhuang et al. [19]. Moreover, the liquidus around the CuCe phase (X_{Ce} = 0.4–0.7) did not agree very well with the experimental data. In this work, the discrepancy was unavoidable since a good conformity of this part would lead to large discrepancies of the invariant reactions over 0–33.3 at.% Ce as shown by the dashed line in Fig. 3. What we did is to ensure the agreement of the invariant reaction temperatures at the low cost of the discrepancy of partial liquidus. Zhuang et al. [19] did the same when they optimized the Cu–Pr system. As is shown in Fig. 5, the calculated enthalpies of mixing of liquid at 1523 K are in good agreement with that of Turchanin et al. [16]. Indicated by the calculated mixing enthalpies of liquid for Cu–RE system [34,35], the minimum values are about 40 at.% RE and our work reflects the same trend. Fig. 6 shows the calculated standard enthalpies of formation of intermetallic compounds. We did not set a large weight on the enthalpy of formation data reported by Fitzner and Kleppa [18] because of the large scatter. Another reason is they are not coordinated with the experimental data on phase diagram. An attempt to lower the enthalpy of formation for Cu_2Ce would result in a rise of temperature for the eutectic reaction $L \leftrightarrow Cu_4Ce + Cu_2Ce$. In this case, it is reasonable as long as the calculation results are within the experimental error. Altogether, the above results manifest the phase diagram and thermodynamic properties can be well described by the set of parameters optimized in this work.

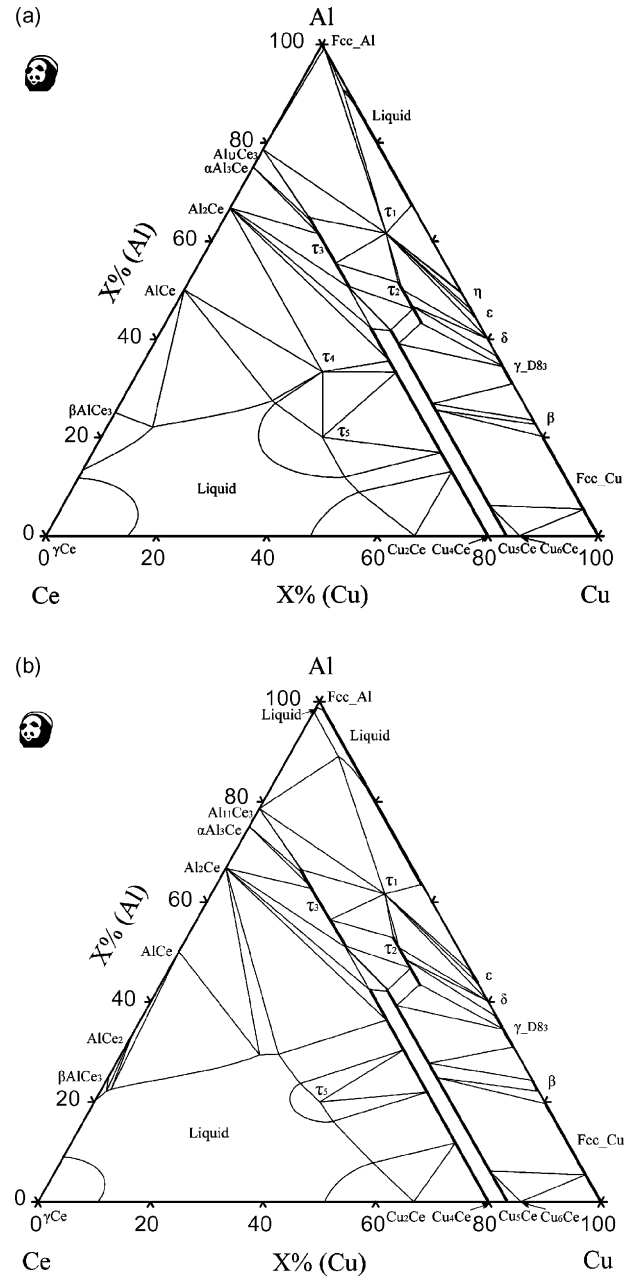

Fig. 12. Calculated isothermal sections at 600 °C (a) and 650 °C (b).

4.2. The Al–Ce–Cu ternary system

On the basis of the experimental information reported by Belov and Khvan [29], the Al-rich part of the Al–Ce–Cu system was optimized. The calculated liquidus projection for this part is shown in Fig. 7 and the calculated invariant reactions are listed in Table 3. Table 4 shows the comparison between the calculated and experimental results regarding the invariant reactions and they agree well with each other. It is worthy of noting that the $Al_{11}Ce_3$ phase in our

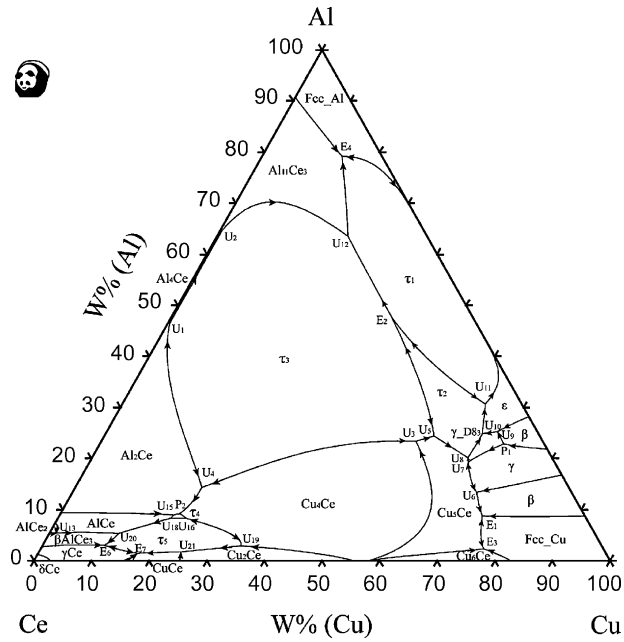

Fig. 13. Calculated liquidus projection of the Al–Ce–Cu ternary system.

results is the Al_4Ce phase in the paper of Belov and Khvan [29] as $Al_{11}Ce_3$ is adopted in the Al–Ce phase diagram for better representation of stoichiometric ratio than Al_4Ce. Two isothermal sections at 540 and 590 °C are also plotted, as shown in Fig. 8. Additionally, there is a quasi-binary eutectic reaction liquid ↔ Fcc_Al + τ_1, as shown in Fig. 9. The calculated temperature is 602 °C. As a validation for the optimization of the Al-rich part, Scheil model was used to simulate the process of non-equilibrium solidification for a series of alloys. The compositions are 0 wt.% Cu and 10 wt.% Ce, 30 wt.% Cu and 1 wt.% Ce, 14 wt.% Cu and 7 wt.% Ce, 5 wt.% Cu and 2.5 wt.% Ce, respectively. The calculated solidification curves about temperature versus fraction of solid are in great accordance with the experimental results of Belov and Khvan [29] obtained from scanning electron microscope (SEM). According to their statements, when the composition was 0 wt.% Cu and 10 wt.% Ce, the Al_4Ce ($Al_{11}Ce_3$) phase showed up as the primary crystal; when the composition was 30 wt.% Cu and 1 wt.% Ce, τ_1-Al_8CeCu_4 phase was the primary crystal; further reduction of the copper and cerium concentrations while keeping the ratio Cu:Ce = 2 led to the decrease in the volume fraction of the binary eutectic. All these results were confirmed in Fig. 10.

The Cu_4Ce, Cu_5Ce, τ_2, τ_3, τ_4 and τ_5 phases in the diagram were optimized on the basis of the literature [27]. For lack of experimental information outside the Al-rich part, they were optimized according to the composition ranges and available phase relations. The calculated isothermal section at 400 °C is shown in Fig. 11. And the calculated homogeneity regions of Cu_4Ce and Cu_5Ce extend to 40.1 and 34.4 at.% Al, respectively, which agree well with the data from literature [27], i.e., 40 and 35 at.% Al. The two literatures [27,29] contradict each other on whether τ_3-$(Al,Cu)_4Ce$ exists in the Al-rich part. Specifically, there is a Al + τ_3 phase region at 500 °C according to literature [27], while it cannot exist inferring from the isothermal sections at 200 and 540 °C given by Belov and Khvan [29]. In order to ensure the same phase relations in the Al-rich corner as reported by Belov and Khvan [29], the sum of Gibbs energies for Fcc_Al and τ_3 should be more positive than that for $Al_{11}Ce_3$ and τ_1. Therefore, the composition ranges for τ_2 and τ_3 were calculated to be 42.6–47.4 at.% Al and 49.5–59.4 at.% Al, respectively, compared with 51.1–55.3 at.% Al and 60–65 at.% Al in literature [27]. More accurate experiments need to be done to ascertain those composition ranges. As to τ_4-AlCeCu, it decomposes at 633 °C through the peritectic reaction liquid + Cu_4Ce + Al_2Ce ↔ τ_4 in this work and the temperature is between 600 and 650 °C as suggested by Chevalier and Bobet [30]. τ_5-$AlCe_2Cu_2$ phase is one of the products of the reaction liquid + Cu_4Ce → τ_5 + τ_4 at 617 °C. However, according to our calculations, it is $(Al,Cu)_4Ce$ (model for Cu_4Ce) that is in the equilibrium instead of $(Al,Cu)_5Ce$ (model for Cu_5Ce) reported by Chevalier and Bobet [30]. This may be ascribed to the weight loss during the melting process of sample preparation. The calculated isothermal sections at 600 and 650 °C are shown in Fig. 12.

The whole liquidus projection was calculated, as shown in Fig. 13. Enlarged portions near the τ_4 phase and Al–Cu and Al–Ce boundary systems are shown in Fig. 14. The calculated invariant reactions and temperatures are summarized in Table 4. Further experimental data are needed to verify the liquidus projection at high Cu and Ce regions.

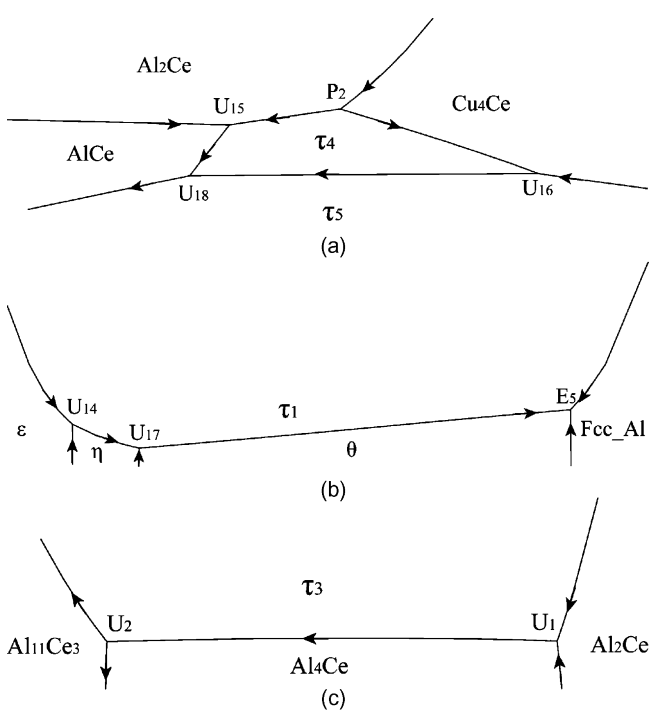

Fig. 14. Enlarged portions of the liquidus projection: (a) near the τ_4 phase; (b) near the Al–Cu boundary system; (c) near the Al–Ce boundary system.

5. Conclusions

The Al–Ce–Cu ternary system was assessed thermodynamically on the basis of reported experimental data and thermodynamic properties for phase diagrams. Good agreement between the calculated and experimental results was obtained and a set of reasonable thermodynamic parameters was now available.

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References

[1] W. Lefebvre, F. Danoix, H. Hallem, B. Forbord, A. Bostel, K. Marthinsen, J. Alloys Compd. 470 (2009) 107–110.

[2] S. Dev, A.A. Stuart, R.C. Ravi Dev Kumaar, B.S. Murty, K. Prasad Rao, Mater. Sci. Eng. A 467 (2007) 132–138.
[3] J. Wloka, S. Virtanen, Acta Mater. 55 (2007) 6666–6672.
[4] Y.T. Li, Z.Y. Liu, Q.K. Xia, Y.B. Liu, Metall. Mater. Trans. A 38 (2007) 2853–2858.
[5] D.H. Xiao, B.Y. Huang, Trans. Nonferrous Met. Soc. China 17 (2007) 1181–1185.
[6] S. Lathabai, P.G. Lloyd, Acta Mater. 50 (2002) 4275–4292.
[7] Z.M. Yin, Q.L. Pan, Y.H. Zhang, F. Jiang, Mater. Sci. Eng. A 280 (2000) 151–155.
[8] N.C. Si, Y. Guo, G.Q. Li, Chin. J. Nonferrous Met. 16 (2006) 606–611.
[9] D.H. Xiao, J.N. Wang, D.Y. Ding, H.L. Yang, J. Alloys Compd. 352 (2003) 84–88.
[10] F. Hanaman, Int. Z. Metallogr. 7 (1915) 174–224.
[11] K.A. Gschneidner Jr., Rare Earth Alloys, D. Van Nostrand Co., Inc., Princeton, NJ, 1961.
[12] U.K. Duisemaliev, A.A. Presnyakov, Zh. Neorg. Khim. 9 (1964) 2258–2260.
[13] T.B. Rhinehammer, D.E. Etter, J.E. Selle, P.Z. Tucker, Trans. Metall. Soc. AIME 230 (1964) 1193–1198.
[14] K.A. Gschneidner Jr., M.E. Verkade, Selected Cerium Phase Diagrams, Report IS-RIC-7, Rare-Earth Information Center, Energy and Mineral Resources Research Institute, Iowa State University, Ames, IA, 18–19, 1974.
[15] A.M. Korolkov, E.V. Lysova, in: A.M. Korolkov (Ed.), Strukt. Svoistva Legk. Splavov, Nauka, Moscow, 1971, pp. 17–20.
[16] M.A. Turchanin, I.V. Nikolaenko, G.I. Batalin, Rasplavy 2 (1988) 25–28.
[17] K. Fitzner, O.J. Kleppa, Metall. Trans. A 24 (1993) 1827–1834.
[18] K. Fitzner, O.J. Kleppa, Metall. Mater. Trans. A 25 (1994) 1495–1500.
[19] W. Zhuang, Z.Y. Qiao, S. Wei, J. Phase Equilib. 17 (1996) 508–521.
[20] W. Dokko, R.G. Bautista, Metall. Trans. B 11 (1980) 511–518.
[21] G.J. Qi, K. Itagaki, A. Yazawa, Mater. Trans. JIM 30 (1989) 273–282.
[22] N. Saunders, I. Ansara, A.T. Dinsdale, M.H. Rand (Eds.), COST-507: Thermodynamic Database for Light Metal Alloys, European Communities, Luxemburg, 1998, pp. 28–33.
[23] V.T. Witusiewicz, U. Hecht, S.G. Fries, S. Rex, J. Alloys Compd. 385 (2004) 133–143.
[24] G. Cacciamani, R. Ferro, CALPHAD 25 (2001) 583–597.
[25] M.C. Gao, et al., Metall. Mater. Trans. A 36 (2005) 3269–3279.
[26] I. Yunusov, I.N. Ganiev, E.A. Shishkin, Izv. Akad. Nauk SSSR Met. 3 (1991) 200–203.
[27] W. Martienssen, Landolt-Börnstein-Group IV Physical Chemistry, Light Metal Systems, Part 1, Springer, Berlin, Heidelberg, 2004, pp. 181–195.
[28] N.A. Belov, A.V. Khvan, A.N. Alabin, Mater. Sci. Forum 519–521 (2006) 395–400.
[29] N.A. Belov, A.V. Khvan, Acta Mater. 55 (2007) 5473–5482.
[30] B. Chevalier, J.-L. Bobet, Intermetallics 9 (2001) 835–838.
[31] A.T. Dinsdale, CALPHAD 15 (1991) 317–425.
[32] B. Sundman, B. Jansson, J.O. Andersson, CALPHAD 9 (1985) 153–190.
[33] PANDAT software for multicomponent phase diagram calculations by CompuTherm, LLC, Madison, WI, since 2000.
[34] L.G. Zhang, G.X. Huang, H.Y. Qi, B.R. Jia, L.B. Liu, Z.P. Jin, J. Alloys Compd. 470 (2009) 214–217.
[35] L.G. Zhang, L.B. Liu, H.S. Liu, Z.P. Jin, CALPHAD 31 (2007) 264–268.